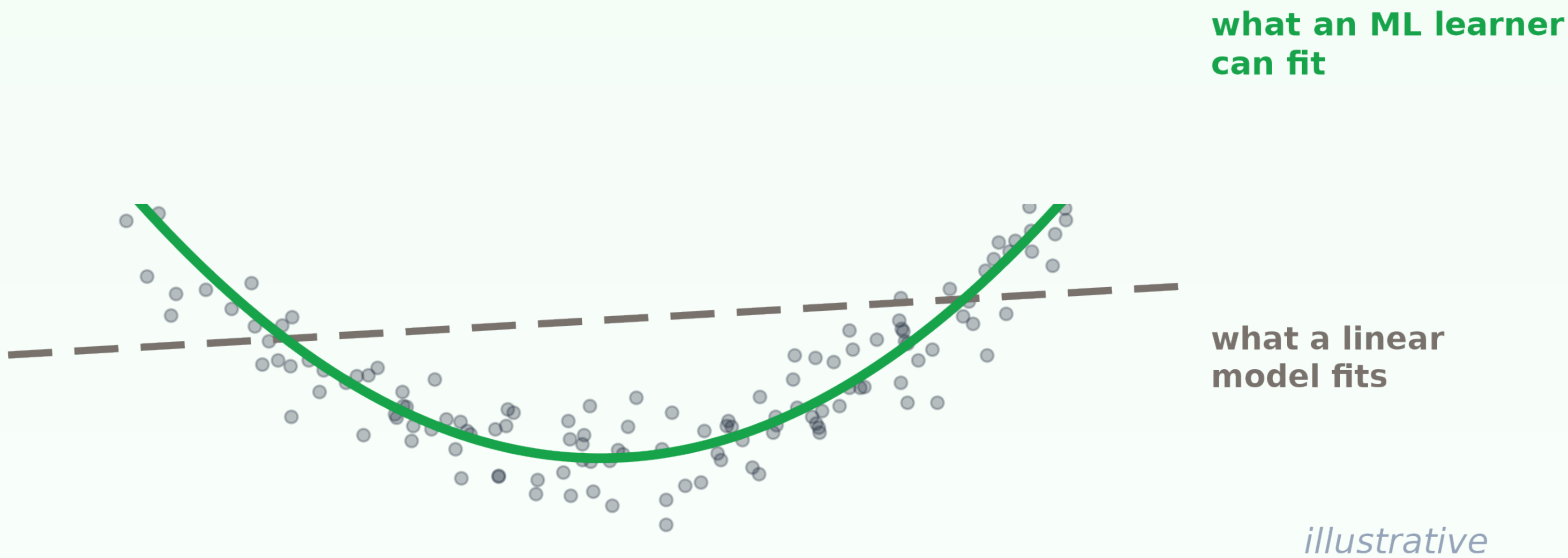
diff - diff

Plug any ML model into your DiD.



New in diff-diff 3.11: the DMLDiD estimator.

Double/debiased machine learning DiD (Chang 2020).

WHY YOU'D WANT THIS

Covariate adjustment is a modeling bet.

Classic doubly-robust DiD (CallawaySantAnna and friends) fixes the nuisance models up front: a logit for treatment, a linear regression for trends.

When treatment or trends bend in X, that bet loses - and nothing in the output tells you.

THE RECEIPT

Confidently wrong.

2.5909

reported SE 0.0743

linear outcome model -

a narrow nominal interval

2.2388

the true effect

simulated staggered panel -

truth known by construction

Almost five standard errors off.

600 units, 6 periods, cohorts treated at t = 4 and 5, 354 never treated. Assignment and trends both bend in X - the linear model cannot. (SEs here are nominal, not theory-backed: the DGP deliberately breaks the rate conditions.)

IT GETS WORSE

Ridge won't save you.

2.5898

reported SE 0.0736 - same miss, same nominal precision

A penalty on the wrong functional form
is still the wrong functional form.

Regularization tunes a model family. It cannot leave the family. (Reported SEs are nominal - illustrative under this DGP.)

THE PAPER

Chang (2020):

stop betting. Learn them.

1 **Neyman-orthogonal scores**

built so small nuisance errors hit the estimate only at second order

2 **Cross-fitting**

nuisances learned on K-1 folds, applied to the held-out fold - flexible learners without overfitting bias

Double/debiased machine learning for difference-in-differences models - The Econometrics Journal, 23(2).

THE MATH

One orthogonal score per cell.

outcome-model residual
(what the trend model missed)

$$s = \left[D - \hat{g}(X) \frac{1 - D}{1 - \hat{g}(X)} \right] \cdot \frac{\Delta Y - \hat{\ell}(X)}{\hat{p}}$$

propensity weight
(treated vs reweighted controls)

treated share
(normalizer)

$\hat{g}(X)$: propensity learner $\hat{\ell}(X)$: outcome learner $\hat{p}$: treated share

Orthogonal. The score's derivative in the LEARNED nuisances g and l is zero at the truth - learner error is second-order. (The treated share p gets its own variance correction.)

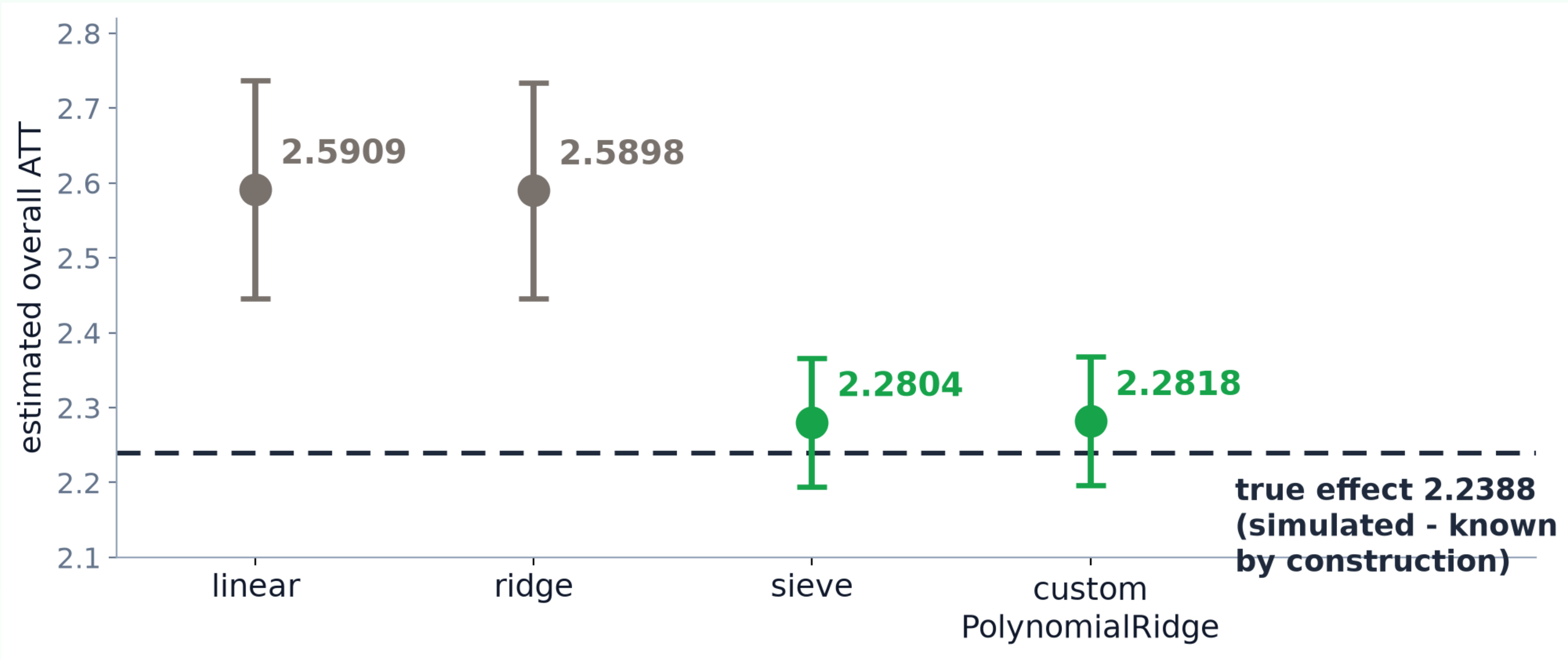
Cross-fit. No unit is scored by a learner that trained on it - flexibility without self-overfitting.

Chang eq. 3.1 (panel case) as implemented: the mean of the summand s over a (cohort, period) cell is that cell's ATT; the centered score is psi = s - ATT.

THE PAYOFF

Same data. Same propensity.

Flexible outcome model: truth.



That is double robustness doing real work: the propensity model is misspecified in every arm - a flexible outcome model alone rescues the point estimate.

Seed-locked simulated example; whiskers are 1.96 SE, illustrative under the deliberately misspecified propensity.

WHEN TO REACH FOR IT

DMLDiD

Covariates with nonlinear or high-dimensional relationships to treatment or trends.

Bring any learner: fit() / predict().

CallawaySantAnna

A logit + linear spec is plausible for your covariates.

Fewer moving parts. Still doubly robust.

Same staggered surface either way: ATT(g,t) cells, event studies, group effects, HonestDiD.

Four learners built in: linear, ridge, sieve, logit - no extra installs.

Or bring your own: scikit-learn estimators already fit the fit() / predict() contract.

THE CODE

Three lines. Any learner.

```
est = DMLDiD(
    outcome_learner="sieve",
    n_folds=5, seed=42,
)
res = est.fit(df, outcome="y", ...,
              covariates=["x1", "x2"])
res.aggregate("event_study")

# or bring your own - sklearn fits the contract:
from sklearn.ensemble import GradientBoostingRegressor
DMLDiD(outcome_learner=GradientBoostingRegressor())
```

seed= pins the fold draws - the shown call reproduces the shown numbers exactly.

BUILT FOR PRODUCTION

The full staggered toolkit.

Post-fit aggregation

event studies, group effects, HonestDiD, uniform sup-t bands

Survey data

designs + clustering: PSU-cohesive folds; design df: 20 PSUs - 4 strata = 16

Repeated cross-sections

panel=False runs Chang's Case 2: fresh samples from the SAME target population each wave

Cross-fit diagnostics

per-cell fold losses, overlap, trimming - inspect every nuisance

Chang Case 1 score vs DoubleML: ATT diff 4.4e-16

committed parity spike; staggered cells + survey lanes are documented extensions of the paper diff - diff

Your covariates aren't linear.

Your DiD can keep up now.

```
pip install diff-diff
```

Tutorial 32 reproduces every number on this deck - learner comparison, HonestDiD, and the survey lane included:

diff-diff.readthedocs.io/en/latest/tutorials/32_dml_did.html

github.com/igerber/diff-diff

Chang (2020), The Econometrics Journal 23(2) - doi:10.1093/ectj/utaa001.